COSC1127/1125 Artificial Intelligence

School of Computing Technologies
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Tutorial No. 9
Reinforcement Learning

You can check this video on $\max/\min$ vs $\arg\max/\arg\min$; and this video on the formulas for Temporal Difference in the AIMIA book.

PART 1: Passive agents . . . .

In the passive *Adaptive Dynamic Programming* (ADP) paradigm, an agent estimates the transition model T using tables of frequencies (model-based learning). Consider the 2 by 3 gridworld with the policy depicted as arrows and the terminal states are illustrated with X's.

	1	2
1	X	←
2	↑	←
3	X	↑

Cells are indexed by (row, column).

The passive ADP agent sees the following state transition observations:

Trial 1: $(3,2) \to (2,2) \to (2,1) \to (1,1)$
Trial 2: $(3,2) \to (2,2) \to (1,2) \to (1,1)$
Trial 3: $(3,2) \to (2,2) \to (2,1) \to (3,1)$
Trial 4: $(3,2) \to (2,2) \to (2,1) \to (2,2) \to (2,1) \to (1,1)$

With perceived rewards as follows:[1]

State	**Reward**
$(3,2)$	-1
$(2,2)$	-2
$(2,1)$	-2
$(1,2)$	-1
$(3,1)$	-10
$(1,1)$	$+16$

(a) Compute the transition model by computing:

[1]This is taking a the convention of rewards given to states (i.e., $R(s)$) and not to transitions (i.e., $R(s,a,s')$), as done in the textbook. As we have discussed in class, both are standard conventions and they can be reduced to each other without loss of generality.

i. the table of state-action frequencies/counts, $N(s, a)$;
ii. the table of state-action-state frequencies/counts, $N(s, a, s')$;
iii. an estimate of the transition model T, using the above tables.

(b) Using the same environment model and observations, now consider how a passive Temporal Difference agent will estimate the utility of the states. The temporal difference learning rule is:

$$U(s) \leftarrow U(s) + \alpha[R(s) + \gamma U(s') - U(s)]$$

If we take $\alpha = 0.5$ as a constant (recall α is the learning rate) and $\gamma = 1$. Compute:
i. the utilities after the first trial;
ii. the utilities after the first and second trials.

(c) **(optional)** Consider the "occasionally-random" and exploration function methods to strike a balance between exploitation and exploration. Recall in the "occasionally-random" approach, $\frac{1}{t}$ of the time the agent selects a random action, otherwise follow the greedy policy. What would a high t value mean? What about a low value t?

Contrast this with the exploration function concept. As an example, consider this exploration function,

$$f(u, n) = \begin{cases} R^+ \text{ , n } < N_e \\ u \text{ , otherwise} \end{cases}$$

What does high/low settings for R^+ and N_e result in?

PART 2: Q-learning . . . .

(a) Consider the grid world from Part 1, but now there are no policies specified. With all Q-values initialised at 0, apply Q-learning taught in class and textbook to learn Q-Values of:
i. all state-action pairs for the first trial/episode;
ii. all state-action pairs for the second trial/episode, given the first.

Recall, the learning rule is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha[R(s) + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

The algorithm in the textbook (and which we follow in this course) does not update the Q-Values of terminal states. Instead, when the next state s' is a terminal state, rather then just setting all previous states, actions and rewards to null, we will additional set all Q-values for that terminal state to its reward. E.g., for state $(1, 1)$, its reward is $+16$, and the first time we visit $(1, 1)$ we will set $Q[(1, 1), N] = Q[(1, 1), S] = Q[(1, 1), E] = Q[(1, 1), W] = +16$.

To simplify calculations, for this question assume actions are deterministic, i.e., if agent goes west, it will go west.

Similar to the TD learning question, use $\alpha = 0.5$ and $\gamma = 1$. For the exploration function, use the one described in Question 1(c), with $R^+ = +10$ and $N_e = 1$. In reality the starting state can vary, but for this question, assume we always start at $(3, 2)$. Note that there can be several answers possible, depending on the action selected when there are tie breakers.

(b) For the following scenarios, briefly describe how we can model them as reinforcement learning problems (states, actions, type of rewards etc).
i. Learning to play chess.
ii. Mary is about to graduate, and she decides to plan her finances for the next 40 years (until retirement). Consider what a reinforcement model might look like for financial planning.

(c) Consider chess. If we wanted to approximate the utility of the states using a sum of weighted features, discuss the type of features that might be useful for playing chess.

End of tutorial worksheet

Exercise 1 continues on the next page. . .